

SOME ELEMENTARY RESULTS

Introduction

The seven sections of this chapter are short and quite elementary. Sections 1, 2, and 5 consist of generalities of a purely logical nature that are used throughout this book. In Section 3 we formally introduce o-minimal structures and prove some simple facts about them. In Section 4 we show that o-minimal ordered groups are abelian and divisible, and that o-minimal ordered rings are real closed fields. In Section 6 we prove the o-minimality of dense linearly ordered sets without endpoints via a characterization of their “definable” sets. In Section 7 we do the same for ordered vector spaces over ordered fields, and consider semilinear sets.

§1. Remarks on logical notation and boolean algebras

(1.1) In this volume we will use logical formulas only in an informal and familiar way, as convenient descriptions or definitions of sets and functions.

(1.2) To illustrate this notational use of formulas, let x, y, z be variables ranging over nonempty sets X, Y, Z respectively, and let $\phi(x, y, z)$ and $\psi(x, y, z)$ be conditions on an arbitrary point $(x, y, z) \in X \times Y \times Z$ defining sets

$$\begin{aligned}\Phi &\subseteq X \times Y \times Z \text{ and } \Psi \subseteq X \times Y \times Z, \\ \Phi &:= \{(x, y, z) \in X \times Y \times Z : \phi(x, y, z) \text{ holds}\}, \\ \Psi &:= \{(x, y, z) \in X \times Y \times Z : \psi(x, y, z) \text{ holds}\}.\end{aligned}$$

Then from the formulas $\phi(x, y, z)$ and $\psi(x, y, z)$ we can construct new formulas:

$\phi(x, y, z) \vee \psi(x, y, z)$, the **disjunction** of $\phi(x, y, z)$ and $\psi(x, y, z)$, defines $\Phi \cup \Psi$;

$\phi(x, y, z) \wedge \psi(x, y, z)$, the **conjunction** of $\phi(x, y, z)$ and $\psi(x, y, z)$, defines $\Phi \cap \Psi$;

$\neg\phi(x, y, z)$, the **negation** of $\phi(x, y, z)$, defines the complement $(X \times Y \times Z) - \Phi$;

$\exists z\phi(x, y, z)$, the **existential quantification** over z of $\phi(x, y, z)$, defines the set

$$\{(x, y) \in X \times Y : \text{there exists } z \in Z \text{ such that } \phi(x, y, z) \text{ holds}\},$$

in other words, $\exists z\phi(x, y, z)$ defines the set $\pi_Z(\Phi)$, where $\pi_Z: X \times Y \times Z \rightarrow X \times Y$ is the obvious projection map; similarly, the formulas $\exists x\phi(x, y, z)$ and $\exists y\phi(x, y, z)$ define the sets $\pi_X(\Phi)$ and $\pi_Y(\Phi)$ where $\pi_X: X \times Y \times Z \rightarrow Y \times Z$ and $\pi_Y: X \times Y \times Z \rightarrow X \times Z$ are projection maps;

$\forall z\phi(x, y, z)$, the **universal quantification** over z of $\phi(x, y, z)$, defines the set

$$\{(x, y) \in X \times Y : \text{for all } z \in Z \text{ the condition } \phi(x, y, z) \text{ holds}\}.$$

Note that this is also the set defined by the formula $\neg \exists z \neg \phi(x, y, z)$.

We write $\phi(x, y, z) \rightarrow \psi(x, y, z)$ as abbreviation for $(\neg \phi(x, y, z)) \vee \psi(x, y, z)$, and $\phi(x, y, z) \leftrightarrow \psi(x, y, z)$ to abbreviate $(\phi(x, y, z) \rightarrow \psi(x, y, z)) \wedge (\psi(x, y, z) \rightarrow \phi(x, y, z))$. Instead of the symbol “ $\wedge$ ” we also use the equivalent symbol “ $\&$ ”.

As an example, the formula $\forall y(\exists z\phi(x, y, z) \rightarrow \exists z\psi(x, y, z))$ defines the set

$$\{x \in X : \text{for all } y \in Y \text{ such that there exists } z \in Z \text{ with } \phi(x, y, z), \\ \text{there exists } z' \in Z \text{ with } \psi(x, y, z')\}.$$

(1.3) Of course, there is nothing special about the use of just three “underlying” sets X, Y, Z . We may as well have more than three, or fewer, and it can happen that $X = Y$, or $Z = X^4$, et cetera. Usually the variables range over the same (nonempty) set, which is often clear from the context and not explicitly specified. The main thing to keep in mind is that the various logical operations on formulas correspond to operations on the sets these formulas define. We shall see in the next two sections that logical notation is more suggestive and transparent than traditional set-theoretic notation, in particular when quantifiers are involved.

(1.4) **DEFINITION.** A **boolean algebra of subsets of a set** X is a nonempty collection $\mathcal{C}$ of subsets of X such that if $A, B \in \mathcal{C}$, then $A \cup B \in \mathcal{C}$ and $X - A \in \mathcal{C}$.

Note that then $X \in \mathcal{C}$ and $\emptyset \in \mathcal{C}$, and that $A, B \in \mathcal{C}$ implies $A \cap B \in \mathcal{C}$.

An **atom** of a boolean algebra $\mathcal{C}$ of subsets of X is a minimal element (with respect to inclusion) of $\{A \in \mathcal{C} : A \neq \emptyset\}$. For example, the atoms of the boolean algebra $\mathcal{P}(X)$ are exactly the singletons $\{x\}$ with $x \in X$. Given sets $A_1, \dots, A_n \subseteq X$, we denote by $B(A_1, \dots, A_n)$ the boolean algebra of subsets of X generated by $A_1, \dots, A_n$, that is, the smallest boolean algebra of subsets of X that contains $A_1, \dots, A_n$ as members. (The notation $B(A_1, \dots, A_n; X)$ would be more correct since it indicates the ambient set X , but X will always be clear from the context.) It is easy to see that the members of $B(A_1, \dots, A_n)$ are exactly the finite unions of sets of the form

$$(*) \quad \left(\bigcap_{i \in \Delta} A_i \right) \cap \left(\bigcap_{j \notin \Delta} (X - A_j) \right), \text{ with } \Delta \subseteq \{1, \dots, n\}.$$

Clearly the nonempty sets of the form $(*)$ are the atoms of $B(A_1, \dots, A_n)$, so $B(A_1, \dots, A_n)$ has at most 2^n atoms, and hence at most 2^{2^n} members.

§2. Elementary facts on structures

(2.1) We define a **structure** on a nonempty set R to be a sequence $\mathcal{S} = (\mathcal{S}_m)_{m \in \mathbb{N}}$ such that for each $m \geq 0$:

- (S1) $\mathcal{S}_m$ is a boolean algebra of subsets of R^m ;
- (S2) if $A \in \mathcal{S}_m$, then $R \times A$ and $A \times R$ belong to $\mathcal{S}_{m+1}$;
- (S3) $\{(x_1, \dots, x_m) \in R^m : x_1 = x_m\} \in \mathcal{S}_m$;
- (S4) if $A \in \mathcal{S}_{m+1}$, then $\pi(A) \in \mathcal{S}_m$, where $\pi : R^{m+1} \rightarrow R^m$ is the projection map on the first m coordinates.

Instead of saying that $\mathcal{S}$ is a structure on R we also say that $(R, \mathcal{S})$ is a structure.

The class of semialgebraic sets mentioned in the Introduction is an example of a structure on $\mathbb{R}$, as will be proved in the next chapter.

Here is another important case. Let Ω be an algebraically closed field, for example, $\Omega = \mathbb{C}$, and let K be a subfield. The **K -constructible subsets of Ω^m** are the finite unions of sets of the form $\{x \in \Omega^m : f_1(x) = \dots = f_k(x) = 0, g(x) \neq 0\}$, where $f_1(X), \dots, f_k(X), g(X) \in K[X]$ and $X = (X_1, \dots, X_m)$. That the K -constructible sets form a structure on Ω is essentially Chevalley's constructibility theorem, also known as quantifier elimination for algebraically closed fields. (In the next volume we can give a very quick proof of this theorem. In this book we only use it as illustration.)

The difficulty in proving in these cases that we are dealing with a structure is the verification of axiom (S4): closure under projections. In some other cases $\mathcal{S}$ is defined in such a way that this axiom is trivially satisfied, but then the difficulty is usually located in verifying closure under complements. (In the next volume we will discuss some general logical ideas—quantifier elimination and model completeness—that are often helpful in dealing with such problems.)

In the following lemmas we collect some elementary facts about a structure $\mathcal{S} = (\mathcal{S}_m)$ on R . First, we fix some terminology: a set $A \subseteq R^m$ is said to **belong to $\mathcal{S}$** (or to **be in $\mathcal{S}$**) if it belongs to $\mathcal{S}_m$; a map $f: A \rightarrow B$ with $A \subseteq R^m$ and $B \subseteq R^n$ is said to belong to $\mathcal{S}$ if its graph $\Gamma(f) \subseteq R^{m+n}$ belongs to $\mathcal{S}_{m+n}$.

(2.2) **LEMMA.**

- (i) If $A \in \mathcal{S}_m$ and $B \in \mathcal{S}_n$, then $A \times B \in \mathcal{S}_{m+n}$.
- (ii) For $1 \leq i < j \leq m$ the diagonal $\Delta_{ij} := \{(x_1, \dots, x_m) \in R^m : x_i = x_j\}$ belongs to $\mathcal{S}$.
- (iii) Let $B \in \mathcal{S}_n$, and let $i(1), \dots, i(n) \in \{1, \dots, m\}$. Then the set $A \subseteq R^m$ defined by the condition

$$(x_1, \dots, x_m) \in A \Leftrightarrow (x_{i(1)}, \dots, x_{i(n)}) \in B$$

belongs to $\mathcal{S}$. (“Permuting and identifying variables are allowed”.)

PROOF. For (i), note that $A \times B = (A \times R^n) \cap (R^m \times B)$, and use (S1) and (S2). For (ii), let $\Delta := \{(x_1, \dots, x_{j-i}) \in R^{j-i} : x_1 = x_{j-i}\}$, and note that $\Delta_{ij} = R^{i-1} \times \Delta \times R^{m-j+1}$, and use (S3) and (i). For (iii) we note that

$$(x_1, \dots, x_m) \in A \Leftrightarrow \exists y_1 \dots \exists y_n (x_{i(1)} = y_1 \ \& \ \dots \ \& \ x_{i(n)} = y_n \ \& \ (y_1, \dots, y_n) \in B).$$

Now think of $(x, y) = (x_1, \dots, x_m, y_1, \dots, y_n)$ as ranging over $R^m \times R^n = R^{m+n}$; the formula “ $(y_1, \dots, y_n) \in B$ ” viewed as a condition on (x, y) defines the set $R^m \times B$ in $\mathcal{S}$, hence by (ii) the formula that follows the quantifiers $\exists y_1 \dots \exists y_n$ defines a set in $\mathcal{S}$. Applying the quantifiers means taking successive images under the projection maps

$$R^{m+n} \rightarrow R^{m+n-1} \rightarrow \dots \rightarrow R^m,$$

hence $A \in \mathcal{S}_m$ by (S4). $\square$

(2.3) LEMMA. *Let $S \subseteq R^m$ and let $f: S \rightarrow R^n$ be a map that belongs to $\mathcal{S}$, that is, $\Gamma(f) \in \mathcal{S}_{m+n}$. Then we have the following properties:*

- (i) $S \in \mathcal{S}_m$,
- (ii) if $A \subseteq S$, $A \in \mathcal{S}_m$, then $f(A) \in \mathcal{S}_n$ and the restriction $f|_A$ belongs to $\mathcal{S}$,
- (iii) if $B \in \mathcal{S}_n$, then $f^{-1}(B) \in \mathcal{S}_m$,
- (iv) if f is injective, its inverse f^{-1} belongs to $\mathcal{S}$,
- (v) if $f(S) \subseteq T \subseteq R^n$, and $g: T \rightarrow R^p$ is a second map belonging to $\mathcal{S}$, then the composition $g \circ f: S \rightarrow R^p$ belongs to $\mathcal{S}$.

PROOF. Let x range over R^m and y over R^n . For (i), use the equivalence $x \in S \Leftrightarrow \exists y(x, y) \in \Gamma(f)$, for (ii) use the equivalence $y \in f(A) \Leftrightarrow \exists x(x \in A \ \& \ (x, y) \in \Gamma(f))$, for (iii) use

$$x \in f^{-1}(B) \Leftrightarrow \exists y(y \in B \ \& \ (x, y) \in \Gamma(f)),$$

for (iv), use part (iii) of the previous lemma, and the equivalence

$$(y, x) \in \Gamma(f^{-1}) \Leftrightarrow (x, y) \in \Gamma(f).$$

For (v), use the previous lemma, and the equivalence (with z ranging over R^p)

$$(x, z) \in \Gamma(g \circ f) \Leftrightarrow \exists y((x, y) \in \Gamma(f) \ \& \ (y, z) \in \Gamma(g)). \quad \square$$

I hope these short arguments illustrate the utility of logical notation and its interpretation in terms of set operations. There are numerous small facts of this kind that the reader can verify instantaneously, by just writing down a suitable logical formula. Below we give some exercises to become familiar with this technique, which is practiced frequently but often implicitly in these notes. The results of these exercises are also used in the rest of the book.

(2.4) Let R be a nonempty set. Given two structures $\mathcal{S}(1)$ and $\mathcal{S}(2)$ on R we write $\mathcal{S}(1) \subseteq \mathcal{S}(2)$ if $\mathcal{S}(1)_m \subseteq \mathcal{S}(2)_m$ for all m ; this defines a partial ordering on

the collection of structures on R . Any family $(\mathcal{S}(i))_{i \in I}$ of structures on R has a greatest lower bound $\mathcal{S}$ in the collection of structures on R , namely

$$\mathcal{S} = \bigcap_i \mathcal{S}(i) \text{ with } \mathcal{S}_m := \bigcap_i \mathcal{S}(i)_m, \text{ for each } m.$$

(2.5) EXERCISES.

1. Let $A \subseteq R^m$ and let $f = (f_1, \dots, f_n) : A \rightarrow R^n$ be a map with component functions $f_i : A \rightarrow R$. Show that f belongs to $\mathcal{S}$ if and only if each f_i belongs to $\mathcal{S}$.
2. (Sheaf property) Let I be a finite index set, let $A \in \mathcal{S}_m$ be the union of the sets $A_i \in \mathcal{S}_m$ ($i \in I$). Show that a map $f : A \rightarrow R^n$ belongs to $\mathcal{S}$ if and only if all its restrictions $f|_{A_i}$ belong to $\mathcal{S}$.
3. Given $A \subseteq R^{m+n}$ and $x \in R^m$ we put $A_x := \{y \in R^n : (x, y) \in A\}$. Show that if $A \in \mathcal{S}_{m+n}$ and $k \in \mathbb{N}$, then the sets $\{x \in R^m : |A_x| \leq k\}$ and $\{x \in R^m : |A_x| = k\}$ belong to $\mathcal{S}_m$.
4. Let the sets A, B, C and the function $f : A \times B \rightarrow C$ belong to $\mathcal{S}$. Show that the set $\{a \in A : f(a, \cdot) : B \rightarrow C \text{ is injective}\}$ belongs to $\mathcal{S}$, and show also that the set $\{a \in A : f(a, \cdot) : B \rightarrow C \text{ is surjective}\}$ belongs to $\mathcal{S}$.
5. Let $P \subseteq R$ be a nonempty subset belonging to $\mathcal{S}_1$. For $m = 0, 1, 2, \dots$ put

$$(\mathcal{S}|P)_m := \{A \cap P^m : A \in \mathcal{S}_m\}, \text{ a boolean algebra of subsets of } P^m.$$

Show that $\mathcal{S}|P := ((\mathcal{S}|P)_m)_{m \in \mathbb{N}}$ is a structure on P . (The “restriction of $\mathcal{S}$ to P ”.)

6. Suppose $\mathcal{S}$ contains binary operations $+: R^2 \rightarrow R$ and $\cdot : R^2 \rightarrow R$ with respect to which R is a ring (always associative with 1 in this book). Show that $\mathcal{S}$ contains $\{0\}$ and $\{1\}$, and that if $\mathcal{S}$ contains $A \subseteq R^m$ and the functions $f, g : A \rightarrow R$, then it contains the functions $-f$, $f + g$, and $f \cdot g$ from A into R .
7. Suppose $R = \mathbb{R}$ and $\mathcal{S}$ contains the order relation $\{(x, y) \in \mathbb{R}^2 : x < y\}$. Show that the topological closure $\text{cl}(A)$ of a set $A \in \mathcal{S}_m$ also belongs to $\mathcal{S}$. Show that if a function $f : \mathbb{R}^{m+1} \rightarrow \mathbb{R}$ belongs to $\mathcal{S}$, then the set

$$A := \{a \in \mathbb{R}^m : f(a, t) \text{ tends to a limit } l(a) \in \mathbb{R} \text{ as } t \rightarrow +\infty\}$$

belongs to $\mathcal{S}$, and the limit function $l : A \rightarrow \mathbb{R}$ so defined belongs to $\mathcal{S}$.

8. Suppose $R = \mathbb{R}$ and $\mathcal{S}$ contains the graphs of addition and multiplication. Show that $\mathcal{S}$ contains the order relation $\{(x, y) \in \mathbb{R}^2 : x < y\}$, and each singleton $\{q\}$ with q a rational number. Show that if $\mathcal{S}$ contains a function $f : I \rightarrow \mathbb{R}$, with open $I \subseteq \mathbb{R}$, then it contains the set $I' := \{x \in I : f \text{ is differentiable at } x\}$, and the derivative $f' : I' \rightarrow \mathbb{R}$.

(2.6) DIGRESSION: THE SMALLEST STRUCTURE ON THE REAL FIELD CONTAINING $\mathbf{Z}$. In the introduction to this book I mentioned the possibility that, starting with some relatively simple sets in euclidean space and performing the usual elementary operations on them, we obtain ever more complicated sets when we iterate this generation process indefinitely. Here is the main example of this phenomenon.

Let $\mathcal{S}$ be the smallest structure on the set $\mathbb{R}$ that contains the graphs of addition and multiplication on $\mathbb{R}$, the subset $\mathbf{Z}$ of $\mathbb{R}$, and each singleton $\{r\}$, $r \in \mathbb{R}$. Then each $\mathcal{S}_n$ consists exactly of the so-called projective subsets of $\mathbb{R}^n$, in particular, all Borel sets in $\mathbb{R}^n$ belong to $\mathcal{S}_n$; see for example Exercise 37.6 in the book *Classical descriptive set theory* by A. Kechris (Springer-Verlag, 1995).

The set $\mathbf{Z}$ of integers can of course be replaced here by the sine function on $\mathbb{R}$, since $\mathbf{Z} = \{a \in \mathbb{R} : \sin(\pi a) = 0\}$.

Those familiar with descriptive set theory and with Matijasevich's theorem on diophantine sets may find it an amusing exercise to show that one can obtain all open subsets of $\mathbb{R}^n$ from a single polynomial equation over $\mathbf{Z}$:

There is a polynomial $P_n(X, Y, Z)$ with integer coefficients in the variables $X = (X_0, X_1, \dots, X_n)$, $Y = (Y_1, \dots, Y_p)$, $Z = (Z_1, \dots, Z_q)$ (for certain $p, q \in \mathbf{N}$ depending on n) such that $\{A_r : r \in \mathbb{R}\}$ is equal to the collection of all open subsets of $\mathbb{R}^n$, where $A \subseteq \mathbb{R}^{n+1}$ is the image of the set

$$\{(x, y, z) \in \mathbb{R}^{n+1+p} \times \mathbf{Z}^q : P_n(x, y, z) = 0\}$$

under the projection map $\mathbb{R}^{n+1+p+q} \rightarrow \mathbb{R}^{n+1}$ onto the first $n+1$ coordinates.

Once all open sets in euclidean spaces are available we also obtain their complements, the closed sets, hence all real-valued continuous functions on closed sets (which have closed graphs), and so on. See Kechris's book referred to earlier for how this generation process continues and never stops.

§3. O-minimal structures

(3.1) NOTATIONS AND CONVENTIONS.

Given linearly ordered sets R_1 and R_2 and a map $f: R_1 \rightarrow R_2$ we say:

- f is **strictly increasing** if $x < y$ in R_1 implies $f(x) < f(y)$ in R_2 ,
- increasing** if $x \leq y$ in R_1 implies $f(x) \leq f(y)$ in R_2 ,
- strictly decreasing** if $x < y$ in R_1 implies $f(x) > f(y)$ in R_2 ,
- decreasing** if $x \leq y$ in R_1 implies $f(x) \geq f(y)$ in R_2 ,
- strictly monotone** if f is either strictly increasing or strictly decreasing,
- monotone** if f is either increasing or decreasing.

A linearly ordered set R is called **dense** if for all $a, b \in R$ with $a < b$ there is $c \in R$ with $a < c < b$. A subset X of a linearly ordered set R is called **convex** (in R) if $a < b < c$ with $a, c \in X$ implies $b \in X$.

Let $(R, <)$ be a dense linearly ordered nonempty set without endpoints. (That $(R, <)$ has no endpoints means that R has no largest or smallest element.)

For convenience, we add two endpoints $-\infty$ and $+\infty$, with $-\infty < a < +\infty$ for all $a \in R$, and put $R_\infty := R \cup \{-\infty, +\infty\}$. An **interval** is always a nonempty “open” interval

$$(a, b) := \{x \in R : a < x < b\} \text{ with } -\infty \leq a < b \leq +\infty.$$

Note that intervals are convex.

Here are some further notations for certain kinds of convex subsets of R :

$$(a, b] := \{x \in R : a < x \leq b\} \text{ where } -\infty \leq a < b < +\infty,$$

$$[a, b) := \{x \in R : a \leq x < b\} \text{ where } -\infty < a < b \leq +\infty,$$

$$[a, b] := \{x \in R : a \leq x \leq b\} \text{ where } -\infty < a \leq b < +\infty.$$

(N.B. Sets of these three kinds will never be referred to as intervals in this book.)

REMARK. We use the same notation for an interval $(a, b) \subseteq R$ as for an ordered pair $(a, b) \in R^2$. It should always be clear from the context which is meant.

We equip R with the **interval topology** (the intervals form a base), and each product R^m with the corresponding product topology, a base of which is formed by the boxes in R^m : a **box** in R^m is a cartesian product $(a_1, b_1) \times \cdots \times (a_m, b_m)$ of intervals. Note that R^m is a Hausdorff space with this topology.

The (topological) closure in R^m of a set $A \subseteq R^m$ is denoted by $\text{cl}(A)$, and its interior (in R^m) by $\text{int}(A)$.

Given functions $f, g: X \rightarrow R_\infty$ on a set $X \subseteq R^m$ we put

$$(f, g) := \{(x, r) \in X \times R : f(x) < r < g(x)\},$$

$$[f, g] := \{(x, r) \in X \times R_\infty : f(x) \leq r \leq g(x)\}.$$

We consider (f, g) as a subset of R^{m+1} ; also $[f, g] \subseteq R^{m+1}$ if f and g are R -valued.

We write $f < g$ to indicate that $f(x) < g(x)$ for all $x \in X$.

(3.2) Let $(R, <)$ be a dense linearly ordered nonempty set without endpoints. An **o-minimal structure on** $(R, <)$ is by definition a structure $\mathcal{S}$ on R such that

$$(O1) \quad \{(x, y) \in R^2 : x < y\} \in \mathcal{S}_2,$$

(O2) the sets in $\mathcal{S}_1$ are exactly the finite unions of intervals and points.

In that case we also say that $(R, <, \mathcal{S})$ is an o-minimal structure.

For the rest of this section we fix an o-minimal structure $\mathcal{S}$ on $(R, <)$.

CONVENTION. Instead of saying that a set A belongs to $\mathcal{S}$ we usually follow more traditional terminology and say that A is **definable**, and similarly with maps. This is of course only permitted if it is clear from the context which $\mathcal{S}$ is meant.

The lemmas that follow are quite simple and will be used frequently without explicit reference.

(3.3) LEMMA. *Let $A \subseteq R$ be definable. Then:*

- (i) *$\inf(A)$ and $\sup(A)$ exist in R_∞ (Dedekind completeness for definable sets),*
- (ii) *the boundary $\text{bd}(A) := \{x \in R : \text{each interval containing } x \text{ intersects both } A \text{ and } R - A\}$ is finite, and if $a_1 < \dots < a_k$ are the points of $\text{bd}(A)$ in order, then each interval (a_i, a_{i+1}) , where $a_0 = -\infty$ and $a_{k+1} = +\infty$, is either part of A or disjoint from A .*

This is almost immediate from the definition of o-minimality. Here are other easy results, the first one a variant of exercise 7 in (2.5) above.

(3.4) LEMMA.

- (i) *If $A \subseteq R^m$ is definable, so are $\text{cl}(A)$ and $\text{int}(A)$.*
- (ii) *If $A \subseteq B \subseteq R^m$ are definable sets, and A is open in B , then there is a definable open $U \subseteq R^m$ with $U \cap B = A$.*

PROOF. For (i), note the following equivalence:

$$\begin{aligned} (x_1, \dots, x_m) \in \text{cl}(A) \\ \iff \\ \forall y_1 \dots \forall y_m \forall z_1 \dots \forall z_m [(y_1 < x_1 < z_1 \ \& \ \dots \ \& \ y_m < x_m < z_m) \rightarrow \\ \exists a_1 \dots \exists a_m (y_1 < a_1 < z_1 \ \& \ \dots \ \& \ y_m < a_m < z_m \ \& \ (a_1, \dots, a_m) \in A)]. \end{aligned}$$

Now use the interpretation of the logical symbols in terms of operations on sets. For (ii), note that one can take for U the union of all boxes in R^m whose intersection with B is contained in A . $\square$

REMARK. Lemma (3.4) uses only axiom (O1) of o-minimality, and not (O2).

Next we introduce a notion of connectedness appropriate for definable sets. In Chapter 3, (2.19), exercise 7 we show that if the underlying ordered set $(R, <)$ of $(R, <, \mathcal{S})$ is the ordered set of real numbers, then this agrees (for definable sets) with the usual notion of connectedness.

(3.5) DEFINITION. A set $X \subseteq R^m$ is called **definably connected** if X is definable and X is not the union of two disjoint nonempty definable open subsets of X .

The verification of the following easy results is left as an exercise.

(3.6) LEMMA.

- (1) *The definably connected subsets of R are the following: the empty set, the intervals, the sets $[a, b]$ with $-\infty < a < b \leq +\infty$, the sets $(a, b]$ with $-\infty \leq a < b < +\infty$, and the sets $[a, b]$ with $-\infty < a \leq b < +\infty$.*
- (2) *The image of a definably connected set $X \subseteq R^m$ under a definable continuous map $f: X \rightarrow R^n$ is definably connected.*
- (3) *If X and Y are definable subsets of R^m , $X \subseteq Y \subseteq \text{cl}(X)$, and X is definably connected, then Y is definably connected.*
- (4) *If X and Y are definably connected subsets of R^m and $X \cap Y \neq \emptyset$, then $X \cup Y$ is definably connected.*

Note the following special case of (2).

If the function $f: [a, b] \rightarrow R$ is definable and continuous, then f assumes all values between $f(a)$ and $f(b)$.

At this point it would be possible to continue directly with Chapters 2, 3, 4, and 5, since logically these depend only on the material just treated.

(3.7) EXERCISE. Let $S \subseteq R^{m+n}$ be definable. Show:

- (i) $\{x \in R^m : S_x \text{ is open}\}$ is definable.
- (ii) $\{(x, y) \in R^{m+n} : y \in \text{int}(S_x)\}$ is definable.

§4. O-minimal ordered groups and rings

(4.1) To formulate the next result we define an **ordered group** to be a group equipped with a linear order that is invariant under left and right multiplication:

$$x < y \Rightarrow zx < zy \text{ and } xz < yz.$$

(So the additive group of reals, and the multiplicative group of positive reals, are ordered groups under the usual ordering, but the multiplicative group of all nonzero reals is not.)

(4.2) PROPOSITION. *Suppose $(R, <, \mathcal{S})$ is an o-minimal structure and $\mathcal{S}$ contains a binary operation $\cdot$ on R , such that $(R, <, \cdot)$ is an ordered group. Then the group $(R, \cdot)$ is abelian, divisible and torsion-free.*

Denoting the unit element of the group by 1 we first prove

LEMMA. *The only definable subsets of R that are also subgroups are $\{1\}$ and R .*

PROOF. Given a definable subgroup G we first show that G is convex: if not, then there are $g \in G$, $r \in R - G$ with $1 < r < g$. This gives a sequence

$$1 < r < g < rg < g^2 < rg^2 < g^3 < \dots$$

whose terms alternate in being in and out of the definable set G , which is impossible. So G is convex, hence assuming $G \neq \{1\}$ we have $s := \sup(G) > 1$ with $(1, s) \subseteq G$. If $s = +\infty$, then clearly $G = R$. If $s < +\infty$, then we take any $g \in (1, s)$, and obtain $s = g \cdot g^{-1}s \in G$, since $g^{-1}s \in (1, s)$, hence $s < gs \in G$, contradicting the definition of s .

PROOF OF PROPOSITION (4.2). For each $r \in R$ the centralizer $C_r := \{x \in R : rx = xr\}$ is a definable subgroup containing r , so $C_r = R$ by the lemma. Hence R is abelian. For each $n > 0$ the subgroup $\{x^n : x \in R\}$ is definable, hence equal to R . This gives divisibility. Every ordered group is torsion-free. $\square$

(4.3) REMARK. Let $(R, <, +)$ be an ordered abelian group, $R \neq \{0\}$, so $(R, <)$ has no endpoints. Assume also that the linearly ordered set $(R, <)$ is dense. Then the addition operation $+: R^2 \rightarrow R$ and the additive inverse operation $-: R \rightarrow R$ are continuous with respect to the interval topology, that is, $(R, +)$ is a topological group with respect to the interval topology.

(4.4) In this book an **ordered ring** is a ring (associative with 1) equipped with a linear order $<$ such that

- (i) $0 < 1$;
- (ii) $<$ is translation invariant, that is: $x < y \Rightarrow x + z < y + z$;
- (iii) $<$ is invariant under multiplication by positive elements:

$$(x < y \text{ and } z > 0 \Rightarrow xz < yz).$$

Note that then the additive group of the ring is an ordered group in the previous sense, that the ring has no zero divisors, that $x^2 \geq 0$ for all x , and that

$$k \mapsto k \cdot 1: \mathbf{Z} \rightarrow \text{ring}$$

is a strictly increasing ring embedding, with the usual ordering on the ring $\mathbf{Z}$ of integers; we shall always identify the ordered ring $\mathbf{Z}$ with its image in the ring under this embedding.

Suppose our ordered ring is moreover a **division ring**: for each $x \neq 0$ there is y with $x \cdot y = 1$. It is easy to check that such a y is unique (written as x^{-1}), and satisfies $y \cdot x = 1$, and that $x > 0$ implies $y > 0$. It is also easy to see that the additive group is divisible, the underlying ordered set is dense without endpoints,

and the maps $(x, y) \mapsto xy$ and $x \mapsto x^{-1}$ (the last map defined only for $x \neq 0$) are continuous with respect to the interval topology.

(4.5) An **ordered field** is an ordered division ring with commutative multiplication. Examples of ordered fields are the field of reals and the field of rational numbers with the usual ordering. In this book we define a **real closed field** to be an ordered field such that if $f(X)$ is a one-variable polynomial with coefficients in the field and $a < b$ are elements in the field with $f(a) < 0 < f(b)$, then there is $c \in (a, b)$ in the field with $f(c) = 0$. (Intermediate value property.) So the ordered field of reals is real closed and the ordered field of rational numbers is not real closed. In the next volume we develop enough model theory so that we can give an efficient and brief treatment of the theory of real closed fields of Artin and Schreier, including Tarski's elimination theory (and much more). Here we will restrict ourselves to the following:

(4.6) **PROPOSITION.** *Suppose $(R, <, \mathcal{S})$ is an $\mathcal{o}$ -minimal structure and $\mathcal{S}$ contains binary operations $+: R^2 \rightarrow R$ and $\cdot: R^2 \rightarrow R$ such that $(R, <, +, \cdot)$ is an ordered ring. Then $(R, <, +, \cdot)$ is a real closed field.*

PROOF. For each $r \in R$ we have a definable additive subgroup rR of $(R, +)$, hence $rR = R$ if $r \neq 0$, by the previous proposition. This shows that $(R, <, +, \cdot)$ is an ordered division ring. Let $\text{Pos}(R) := \{r \in R : r > 0\}$. Clearly $\text{Pos}(R)$ is an ordered multiplicative group. By restricting $\mathcal{S}$ to $\text{Pos}(R)$ (see (2.5), exercise 5) it follows from the previous proposition that multiplication is commutative on $\text{Pos}(R)$, hence on all of R . So $(R, <, +, \cdot)$ is an ordered field. By exercise 6 of (2.5), and the remarks in (4.3) and (4.4), each one-variable polynomial $f(X) \in R[X]$ gives rise to a definable continuous function $x \mapsto f(x): R \rightarrow R$. Now apply the remark at the end of (3.6) to this function. $\square$

§5. Model-theoretic structures

(5.1) A structure $\mathcal{S}$ on a set R is often introduced by indicating relations and functions on R that “generate” $\mathcal{S}$. For example, in the next chapter we will see that the structure on $\mathbb{R}$ generated by its addition and multiplication operations, and the individual real numbers, consists exactly of the semialgebraic sets. It is quite useful also for theoretical reasons to have available a notion of structure that singles out particular relations and functions as “primitives”.

(5.2) **DEFINITION.** A **model-theoretic structure** $\mathcal{R} = (R, (S_i)_{i \in I}, (f_j)_{j \in J})$ consists of a nonempty set R , relations $S_i \subseteq R^{m(i)}$ ($i \in I, m(i) \in \mathbb{N}$), and functions $f_j: R^{n(j)} \rightarrow R$ ($j \in J, n(j) \in \mathbb{N}$). If $n(j) = 0$, we identify f_j with its unique value in R , and call f_j a **constant**. We also call R the **underlying set** of $\mathcal{R}$, and the S_i and f_j the **basic** (or **primitive**) relations and functions of $\mathcal{R}$.

If the index sets I and J are finite we usually just list the relations and functions.

For example, an abelian group will be considered as a model-theoretic structure of the form $(A, 0, -, +)$, with 0 the zero element, $- : A \rightarrow A$ the operation of taking the negative and $+ : A^2 \rightarrow A$ the group operation; an ordered ring is a model-theoretic structure of the form $(R, <, 0, 1, -, +, \cdot)$, et cetera.

Given a model-theoretic structure $\mathcal{R} = (R, (S_i)_{i \in I}, (f_j)_{j \in J})$, and set $C \subseteq R$ we define a new model-theoretic structure $\mathcal{R}_C := (R, (S_i)_{i \in I}, (f_j)_{j \in J}, (c)_{c \in C})$ which has the same underlying set and the same basic relations as $\mathcal{R}$, but has in addition to the basic functions f_j a constant c for each element $c \in C$, where formally we identify c with the corresponding function $R^0 \rightarrow R$.

(5.3) Given a model-theoretic structure $\mathcal{R} = (R, (S_i), (f_j))$, we let $\text{Def}(\mathcal{R})$ be the smallest structure on the set R (in the sense of Section 2) that contains each relation S_i and each function f_j . Note that for each set $C \subseteq R$ the structure $\text{Def}(\mathcal{R}_C)$ on R is larger than or equal to $\text{Def}(\mathcal{R})$. (We can recover $\text{Def}(\mathcal{R}_C)$ from $\text{Def}(\mathcal{R})$ and C in a simple way, see (5.9), exercise 1 below.) In this book we mostly work with $\text{Def}(\mathcal{R}_R)$. The sets $S \subseteq R^m$ and maps $f : S \rightarrow R^n$ that belong to $\text{Def}(\mathcal{R})$ are also called **definable in $\mathcal{R}$** (or just definable, when $\mathcal{R}$ is clear from the context). A point $a = (a_1, \dots, a_m) \in R^m$ is said to be **definable in $\mathcal{R}$** if the set $\{a\} \subseteq R^m$ is definable in $\mathcal{R}$.

We usually write “definable in $\mathcal{R}$ using constants from C ”, or “definable in $\mathcal{R}$ with parameters from C ” in place of “definable in $\mathcal{R}_C$ ”. For $C = R$ we write “definable in $\mathcal{R}$ using constants” or “definable in $\mathcal{R}$ with parameters” instead of “definable in $\mathcal{R}_R$ ”. We often omit “in $\mathcal{R}$ ”, if $\mathcal{R}$ is clear from the context.

(5.4) To discuss in a uniform way all model-theoretic structures of a certain type, for instance all ordered abelian groups, or all real closed fields, it is convenient to have available the notions of a (first-order) language L and L -structure.

A **language L** is a disjoint union of two sets, a set of **relation symbols**, and a set of **function symbols**, each relation symbol S and each function symbol f being equipped with a number: $\text{arity}(S), \text{arity}(f) \in \mathbb{N}$. If $\text{arity}(S) = m, \text{arity}(f) = n$, we also say that S is an m -ary relation symbol, and f is an n -ary function symbol. Function symbols of arity 0 are also called **constant symbols**, and instead of saying “1-ary” and “2-ary” we say “unary” and “binary”.

(5.5) An **L -structure** is a model-theoretic structure $\mathcal{R} = (R, (S^{\mathcal{R}}), (f^{\mathcal{R}}))$ such that for each m -ary relation symbol S and each n -ary function symbol f of L we have

$$S^{\mathcal{R}} \subseteq R^m, \quad f^{\mathcal{R}} : R^n \rightarrow R;$$

$S^{\mathcal{R}}$ and $f^{\mathcal{R}}$ are the **interpretations** of the symbols S and f in $\mathcal{R}$. So the set of relation symbols of L indexes the primitive relations of $\mathcal{R}$, and the set of function symbols of L indexes the primitive functions of $\mathcal{R}$. We usually drop the superscript $\mathcal{R}$ in $S^{\mathcal{R}}$ and $f^{\mathcal{R}}$, leaving it to the context whether the symbol or its interpretation is meant.

(5.6) EXAMPLES.

- (1) The language of abelian groups is $L_{ab} := \{0, -, +\}$: it has a constant symbol 0, unary function symbol $-$, and binary function symbol $+$. We consider abelian groups as L_{ab} -structures in the obvious way.
- (2) The language $L_{ab}(<) := \{<, 0, -, +\}$ of ordered abelian groups has in addition a binary relation symbol $<$, and we regard ordered abelian groups as $L_{ab}(<)$ -structures in the obvious way.
- (3) The language of rings is $L_{ring} := \{0, 1, -, +, \cdot\}$: it extends the language of abelian groups by an extra constant symbol 1 and an extra binary function symbol $\cdot$. We consider rings as L_{ring} -structures in the obvious way.

(In a later volume we develop enough model theory to make better use of the notion of L -structure. In this book it is only a matter of convenient terminology.)

(5.7) DEFINITION. A model-theoretic structure $\mathcal{R} = (R, <, \dots)$, where $<$ is a dense linear order without endpoints on R , is called **o-minimal** if $\text{Def}(\mathcal{R}_R)$ is an o-minimal structure on $(R, <)$, in other words, every set $S \subseteq R$ that is definable in $\mathcal{R}$ using constants is a union of finitely many intervals and points.

(5.8) EXPANSIONS. The simplest thing is just to say how we use this notion in the few places that it occurs in this book. Let $(R, <, \mathcal{S})$ be an o-minimal structure. We say that $(R, <, \mathcal{S})$ **expands an ordered abelian group** if there are abelian group operations $0: R^0 \rightarrow R$, $-: R \rightarrow R$, and $+: R^2 \rightarrow R$ belonging to $\mathcal{S}$ such that $(R, <, 0, -, +)$ is an ordered abelian group; in that case we also say that $(R, <, \mathcal{S})$ **expands the ordered abelian group** $(R, <, 0, -, +)$. In particular, the underlying ordered set of $(R, <, \mathcal{S})$ is the same as the underlying ordered set of the ordered abelian group that $(R, <, \mathcal{S})$ expands. Similarly, we define what it means for $(R, <, \mathcal{S})$ to expand an ordered ring. Note that by the previous section, if the o-minimal structure $(R, <, \mathcal{S})$ expands a certain ordered ring, then this ordered ring is actually a real closed field.

(5.9) EXERCISES.

1. Let $\mathcal{R} = (R, \dots)$ be a model-theoretic structure, $C \subseteq R$, and $A \subseteq R^n$. Show that A is definable in $\mathcal{R}$ using constants from C if and only if there exist a set $S \subseteq R^{m+n}$ definable in $\mathcal{R}$, and elements $c_1, \dots, c_m \in C$ such that for all $(x_1, \dots, x_n) \in R^n$

$$(x_1, \dots, x_n) \in A \Leftrightarrow (c_1, \dots, c_m, x_1, \dots, x_n) \in S.$$

2. Let $\mathcal{R} = (R, \dots)$ be a model-theoretic structure and $S \subseteq R^{m+n}$ definable in $\mathcal{R}$. Show that if $a \in R^m$ is definable in $\mathcal{R}$, then the set $S_a := \{b \in R^n : (a, b) \in S\}$ is definable in $\mathcal{R}$.

3. Let $\mathcal{R} = (\mathbb{R}, <, 0, 1, -, +, \cdot)$ be the ordered field of real numbers. Show that each function $f: \mathbb{R}^m \rightarrow \mathbb{R}$ defined by a polynomial $f(X_1, \dots, X_m) \in \mathbb{R}[X_1, \dots, X_m]$

is definable in $\mathcal{R}$ using constants. Derive that each semialgebraic set in $\mathbb{R}^m$ (see “Introduction and Overview”) is definable in $\mathcal{R}$ using constants.

§6. The simplest o-minimal structures

(6.1) Let $(R, <)$ be a dense linearly ordered nonempty set without endpoints.

We prove below that the model-theoretic structure $(R, <)$ is o-minimal, by describing explicitly all definable sets. These o-minimal structures are too simple to be of much interest, but we treat this case to illustrate the notions of the previous sections. Starting with Chapter 3 we develop systematically the general theory of o-minimal structures. However, in Chapters 6–10 we will assume that our o-minimal structures expand ordered abelian groups or even ordered rings. Such extra assumptions are satisfied in all cases of real interest to us, but not by $(R, <)$, discussed in the present section. In this connection $(R, <)$ can serve as a useful counterexample to potential generalizations of various theorems in these later chapters, showing that the extra assumptions in Chapters 6–10 cannot be omitted.

(6.2) Let $1 \leq i \leq m$. The function $(x_1, \dots, x_m) \mapsto x_i : R^m \rightarrow R$ will be denoted just by x_i . The **simple** functions on R^m are by definition these coordinate functions $x_1, \dots, x_m$, and the constant functions $R^m \rightarrow R$.

(6.3) Let $f_1, \dots, f_N$ be simple functions on R^m , and let $\epsilon : \{1, \dots, N\}^2 \rightarrow \{-1, 0, 1\}$ be given. Then we put

$$\epsilon(f_1, \dots, f_N) := \{x \in R^m : \text{for all } (i, j) \in \{1, \dots, N\}^2 \text{ we have}$$

$$f_i(x) < f_j(x) \text{ if } \epsilon(i, j) = -1,$$

$$f_i(x) = f_j(x) \text{ if } \epsilon(i, j) = 0,$$

$$f_i(x) > f_j(x) \text{ if } \epsilon(i, j) = 1\}.$$

Of course $\epsilon(f_1, \dots, f_N)$ may be empty, for instance when $\epsilon(i, j) \neq -\epsilon(j, i)$ for some pair (i, j) . Suppose that $\epsilon(f_1, \dots, f_N)$ is nonempty. Note that if ξ and η are the restrictions of f_i and f_j to $\epsilon(f_1, \dots, f_N)$, then either $\xi < \eta$, or $\xi = \eta$, or $\eta < \xi$. Let $\xi_1 < \dots < \xi_k$ be the restrictions of $f_1, \dots, f_N$ to $\epsilon(f_1, \dots, f_N)$ arranged in increasing order. One checks easily that the sets $\Gamma(\xi_j)$ ($1 \leq j \leq k$) and the sets (ξ_j, ξ_{j+1}) ($0 \leq j \leq k$, where $\xi_0 = -\infty$ and $\xi_{k+1} = +\infty$ by convention) are exactly the nonempty subsets of R^{m+1} of the form $\epsilon'(f_1, \dots, f_N, x_{m+1})$, where

$$\epsilon' : \{1, \dots, N, N+1\}^2 \rightarrow \{-1, 0, 1\}$$

is an extension of ϵ . (Here a simple function f on R^m is regarded as a simple function on R^{m+1} by setting $f(x_1, \dots, x_m, x_{m+1}) := f(x_1, \dots, x_m)$.) Define a **simple set** in R^m to be a subset of R^m of the form $\epsilon(f_1, \dots, f_N)$ with $f_1, \dots, f_N$ simple functions on R^m and $\epsilon : \{1, \dots, N\}^2 \rightarrow \{-1, 0, 1\}$. We have just proved that if $S \subseteq R^{m+1}$ is simple, then its image under the projection map

$$(x_1, \dots, x_m, x_{m+1}) \mapsto (x_1, \dots, x_m) : R^{m+1} \rightarrow R^m$$

is simple in R^m . We now easily derive

(6.4) PROPOSITION. *The subsets of R^m that are definable in $(R, <)$ using constants are exactly the finite unions of simple sets in R^m .*

PROOF. Let $\mathcal{S}_m$ be the collection of finite unions of simple sets in R^m . Clearly $\mathcal{S}_m$ is a boolean algebra of subsets of R^m , and each set in $\mathcal{S}_m$ is definable in $(R, <)$ using constants. The considerations in (6.3) above show that $\mathcal{S} := (\mathcal{S}_m)_{m \in \mathbb{N}}$ is a structure on the set R , hence the sets in $\mathcal{S}_m$ are exactly the subsets of R^m definable in $(R, <)$ using constants. $\square$

REMARK. It follows easily that $\mathcal{S}$ is an o-minimal structure on $(R, <)$.

(6.5) COROLLARY. *The model-theoretic structure $(R, <)$ is o-minimal.*

(6.6) From a geometric viewpoint $(R, <)$ can have rather bad properties. For instance, given any cardinal κ one can construct a dense linearly ordered nonempty set $(R, <)$ without endpoints and a collection $\mathcal{C}$ of κ intervals in $(R, <)$ such that $|I| \neq |J|$ for distinct $I, J \in \mathcal{C}$. We leave the construction of such an $(R, <)$ and collection $\mathcal{C}$ of intervals to the reader as an exercise, and note only the extreme lack of homogeneity of such an $(R, <)$ for large κ : $\mathcal{C}$ is a collection of κ intervals no two of which are homeomorphic. In Chapters 6–10 we impose extra algebraic structure that implies homogeneity.

§7. Semilinear sets

(7.1) In this section we show that the sets definable using constants in an ordered vector space over an ordered field are exactly the semilinear sets. Ordered vector spaces occur naturally in a variety of situations, but in this book the material of this section will only play a minor role.

(7.2) DEFINITIONS. *Let F denote an ordered field, until further notice.*

An **ordered F -linear space** is a vector space R over F equipped with a linear order making R an ordered additive group, such that for all $\lambda \in F$ and $x \in R$

$$(\lambda > 0, x > 0) \Rightarrow \lambda x > 0.$$

Instead of “ordered F -linear space” we also say “ordered vector space over F ”. Of course, F considered as a vector space over itself, with its usual order, is an ordered F -linear space. More generally, any ordered field extension of F is an ordered F -linear space in an obvious way. Note also that for $F = \mathbb{Q}$ the ordered $\mathbb{Q}$ -linear spaces are exactly the divisible ordered abelian groups, where the latter are made into vector spaces over $\mathbb{Q}$ in the obvious way.

(7.3) *Let R be an ordered F -linear space. We assume $R \neq \{0\}$, so that $(R, <)$ is a dense linearly ordered set without endpoints.*

An **affine function on R^m** is a function $f: R^m \rightarrow R$ of the form

$$f(x_1, \dots, x_m) = \lambda_1 x_1 + \dots + \lambda_m x_m + a,$$

where the $\lambda_i \in F$ and $a \in R$ are fixed. A **basic semilinear set in R^m** is a set of the form

$$\{x \in R^m : f_1(x) = \dots = f_p(x) = 0, g_1(x) > 0, \dots, g_q(x) > 0\},$$

with the f_i and g_j affine functions on R^m . (We allow of course $p = 0$ or $q = 0$.) A **semilinear set in R^m** is a finite union of basic semilinear sets in R^m . (Perhaps “semiaffine” would be a better term than “semilinear”, but “semilinear” seems more popular.)

Note that the intersection of two basic semilinear sets in R^m is a basic semilinear set in R^m . The basic semilinear sets in $R = R^1$ are exactly the intervals, the points, and the empty set. The complement in R^m of a basic semilinear set in R^m is a semilinear set in R^m , but not necessarily a basic semilinear set in R^m .

(7.4) PROPOSITION. *Let $f_1, \dots, f_N$ be affine functions on R^{m+1} . Then we can partition R^m into basic semilinear sets $C_1, \dots, C_k$ such that for each $C \in \{C_1, \dots, C_k\}$ there are functions $\xi_{C,1} < \dots < \xi_{C,j(C)}$ from C into R with the following properties:*

- (i) *each $\xi_{C,j}$ is the restriction of an affine function on R^m ;*
- (ii) *each function f_n ($1 \leq n \leq N$) has constant sign on each of the sets $\Gamma(\xi_{C,j})$ ($1 \leq j \leq j(C)$), and on each of the sets $(\xi_{C,j}, \xi_{C,j+1})$ ($0 \leq j \leq j(C)$, where $\xi_{C,0} := -\infty$, $\xi_{C,j(C)+1} := +\infty$ by convention).*

PROOF. Let $f_1, \dots, f_M$ depend on the last variable x_{m+1} , while the f_n for $M < n \leq N$ do not. The latter may therefore be viewed also as functions on R^m . Write $f_p(x_1, \dots, x_m, x_{m+1}) = \lambda_p(g_p(x_1, \dots, x_m) - x_{m+1})$ for $1 \leq p \leq M$, where $\lambda_p \in F^\times$ and g_p is an affine function on R^m . Take a partition of R^m into basic semilinear sets $C_1, \dots, C_k$ on each of which each function in

$$\{f_{M+1}, \dots, f_N\} \cup \{g_p - g_q : 1 \leq p, q \leq M\}$$

takes constant sign. For each $C \in \{C_1, \dots, C_k\}$, let $\xi_{C,1} < \dots < \xi_{C,j(C)}$ be the restrictions of the functions g_p to C arranged in increasing order. This is possible because the $g_p - g_q$ are of constant sign on C . Then the desired result holds. $\square$

(7.5) REMARK. Note that each of the sets in (ii) is itself a basic semilinear set. The actual construction in the proof produces sets as in (ii) that are of the form

$$\{(x, t) \in C \times R : \text{sign}(f_n(x, t)) = \epsilon(n) \text{ for } n = 1, \dots, N\}$$

for a suitable sign description $\epsilon: \{1, \dots, N\} \rightarrow \{-1, 0, 1\}$. This property is somewhat stronger than (ii).

(7.6) COROLLARY. *Let $\mathcal{S}_m$ be the boolean algebra of semilinear subsets of R^m . Then $\mathcal{S} := (\mathcal{S}_m)_{m \in \mathbb{N}}$ is an o-minimal structure on the ordered set $(R, <)$. Each function in $\mathcal{S}$ is piecewise affine, more precisely, given a function $f: A \rightarrow R$ in $\mathcal{S}$ with $A \subseteq R^m$, there is a partition of A into basic semilinear sets A_i ($1 \leq i \leq k$) such that $f|_{A_i}$ is the restriction to A_i of an affine function on R^m , for each $i \in \{1, \dots, k\}$.*

PROOF. Let $\pi: R^{m+1} \rightarrow R^m$ be the projection map onto the first m coordinates. Given a set $S \in \mathcal{S}_{m+1}$, we have to show that $\pi(S) \in \mathcal{S}_m$. Let $f_1, \dots, f_N$ be the affine functions involved in a description of S as a union of basic semilinear sets. Applying the proposition above to $f_1, \dots, f_N$ we see that S is a union of sets as described in (ii). Since each of the sets in (ii) has as its π -image a basic semilinear set in R^m , it follows that $\pi(S) \in \mathcal{S}_m$. This shows that $\mathcal{S}$ is indeed an o-minimal structure on $(R, <)$. If S as above is the graph of a function $f: A \rightarrow R$, $A \subseteq R^m$, then the sets of type (ii) that are part of S must be of the form $\Gamma(\xi)$, with ξ a restriction of an affine function. Hence f is piecewise affine. $\square$

(7.7) Construe R as a model-theoretic structure for the language

$$L_F := \{<, 0, -, +\} \cup \{\lambda \cdot : \lambda \in F\}$$

of ordered abelian groups augmented by a unary function symbol $\lambda \cdot$ for each $\lambda \in F$, to be interpreted as multiplication by the scalar λ .

(7.8) COROLLARY. *The subsets of R^m definable in the L_F -structure R using constants are exactly the semilinear sets in R^m .*

PROOF. We leave it to the reader to derive this from (7.6). $\square$

(7.9) REMARK. The proofs above are “uniform in R ”, and one particular consequence of this uniformity is that if R' is an ordered F -linear space extending R , then an “elementary statement with constants from R ” holds in R if and only if it holds in R' . Since F can be embedded as an ordered F -linear space in each nontrivial ordered F -linear space, it follows that all nontrivial ordered F -linear spaces satisfy exactly the same elementary statements formulated in the language L_F . These issues will be rigorously treated in a later volume where the necessary logical preliminaries are put in place. (For instance, we have to define “elementary statement”.) In this book we mainly deal with the situation that $R = F$, and we then refer to the semilinear sets in R^m for $m = 0, 1, 2, \dots$ as semilinear sets over R .

(7.10) We never used the commutativity of the multiplication on F : everything in this section goes through when F is an ordered division ring. (“Vector space over the division ring F ” is then interpreted as “left F -module”.) Noncommutative ordered division rings can be obtained from noncommutative ordered groups via a power series construction, see for example Fuchs [25].

(7.11) An ordered vector space R is homogeneous in the sense that for each two points $p, q \in R$ there is a definable homeomorphism from R onto itself that sends p

to q , namely $x \mapsto x + (q - p): R \rightarrow R$. This homogeneity does not necessarily extend from points to intervals, as the following example shows.

EXAMPLE. Equip the cartesian product $\mathbb{R} \times \mathbb{Q}$ of additive groups with the lexicographic ordering:

$$(r_1, q_1) < (r_2, q_2) \text{ iff } r_1 < r_2, \text{ or } r_1 = r_2 \text{ and } q_1 < q_2.$$

This makes $\mathbb{R} \times \mathbb{Q}$ into a divisible ordered abelian group in which there are both countable intervals and uncountable intervals. In particular, not every two intervals of $\mathbb{R} \times \mathbb{Q}$ are homeomorphic, let alone definably homeomorphic. This example is used later to show that certain results on o-minimal structures expanding ordered fields do not hold in general for o-minimal structures expanding ordered abelian groups.

(7.12) DIGRESSION: THE DIVISION RING OF DEFINABLE ADDITIVE MAPS. We have just seen that ordered vector spaces over ordered fields, and, more generally, over ordered division rings, are o-minimal. Here we show, using a result from Chapter 3, that any o-minimal structure that expands an ordered abelian group is in fact an ordered vector space over an ordered division ring that is canonically associated to the structure.

Let $(R, <, \mathcal{S})$ be an o-minimal structure which expands an ordered abelian group $(R, <, 0, -, +)$.

Let $\lambda: R \rightarrow R$ be a definable additive map. (Additive: $\lambda(x + y) = \lambda(x) + \lambda(y)$.) Since λ is definable, the monotonicity theorem from Chapter 3 implies that λ is continuous at some point of R . Then additivity of λ implies that λ is continuous on all of R . Since R has no definable subgroups besides $\{0\}$ and itself it follows by looking at the kernel and the image of λ that either $\lambda = 0$ or λ is a bijection. Suppose λ is a bijection. Then $(0, \infty)$ is the disjoint union of the two definable open sets $\{x > 0 : \lambda(x) > 0\}$ and $\{x > 0 : \lambda(x) < 0\}$, so one of these two sets is all of $(0, \infty)$. In other words, either $\lambda(x) > 0$ for all $x > 0$, or $\lambda(x) < 0$ for all $x > 0$. Write $\lambda > 0$ in the first case and $\lambda < 0$ in the second case. The set of definable additive maps on R is a ring under pointwise addition and composition with identity 1_R , and we have just defined an ordering on this ring. We have shown

PROPOSITION. *The ring of definable additive maps on R is an ordered division ring.*

Hence R is naturally an ordered vector space over this division ring, and the sets definable in this ordered vector space using constants are clearly also definable in $(R, <, \mathcal{S})$. There is more to say in this direction, see [21, §4].

EXERCISE.

Let R be a nontrivial ordered vector space over an ordered division ring F , and consider R as an L_F -structure. Show that the maps $R \rightarrow R$ that are additive and definable using constants are exactly the scalar multiplications by elements of F . (Hence these maps are actually definable without using constants.)

Notes and comments

The results in Section 4 on o-minimal ordered groups and rings are from Pillay and Steinhorn [49]. (There are other notions of “ordered ring” in use that allow nonzero nilpotents, cf. Fuchs [25].) The content of Section 6 is also known among model-theorists as “elimination of quantifiers” for dense linearly ordered sets, cf. Chang & Keisler [9]. The key argument in the proof of Proposition 7.4 is a form of “Fourier-Motzkin elimination”, and thus goes back a long time. See also Sontag [55].

The results in sections 4, 6 and 7 already hint towards a division of all o-minimal structures into three fundamentally different kinds:

- (1) Those in which very few sets are definable, like the dense linearly ordered sets without endpoints.
- (2) Non-trivial ordered vector spaces over ordered division rings.
- (3) O-minimal expansions of real closed fields.

A precise result along these lines, the Trichotomy Theorem for o-minimal structures, has indeed been established by Peterzil and Starchenko [46]. This theorem can be seen as a manifestation of the so-called Zil’ber Principle of geometric model-theory, which clarifies the intrinsic nature of such classifications. These matters fall unfortunately outside the scope of this book.

